

An Analysis-Guided Workflow for Constructing Playable Csound Physical Models: A Tupipa Case Study

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Abstract. The Tupipa is a traditional three-string plucked lute from southwest China whose sound is preserved mainly through recordings and documentation. Its three strings form two courses: a double course tuned in unison and a single-string course, the latter providing a controlled first case for playable modeling. Such materials document performance practice but do not provide playable controls for physical-model synthesis. This paper presents an analysis-guided workflow for constructing playable Csound physical models from recordings, supplemented by limited physical measurements. The workflow combines ATS and time-domain analysis of recordings with rule-based parameter initialization, diagnostic acoustic-descriptor comparison, bounded local parameter search, and listening-panel calibration. The case study therefore focuses on the single-string course and links measured plucking position, ATS F0, partial balance, decay, and residual/noise cues to Csound controls. In the model-construction stage, initialized controls are instantiated as a four-layer Csound instrument: a `wgpluck2` core string-vibration layer, a plectrum-contact layer, a friction/noise layer, and a body-response layer. Rather than introducing another instrument-specific synthesizer, the study shows how recordings can become playable, physically interpretable Csound instruments while retaining listening judgment in the final calibration stage.

Keywords: Csound, physical modeling, Tupipa, descriptor-to-parameter mapping, plucked-string synthesis

1 Introduction

The Tupipa is a traditional three-string plucked lute from southwest China that is central to the performance of Nanping Quzi, a Chinese national intangible cultural heritage tradition [1]. Similar to the modern Pipa, it belongs to the broader lute family, but its three strings form two courses: a double course tuned in unison and a single course [2]. The double course introduces coupled-string behavior that is outside the scope of the present study, so the single-string course is used here as a controlled first step toward a complete playable instrument.

Digital preservation commonly relies on sound archives, video documentation, photography, 3D scanning, and database records. These materials are valuable for access and documentation, but Csound-based composition and performance require a further technical decision: how descriptors extracted from recordings can guide the initialization and calibration of playable synthesis controls. Spectral methods such as ATS characterize sinusoidal partials and residual/noise energy, while physical models expose controls related to pitch, excitation position, damping, and resonance behavior [3,4,5,6]. The descriptor-to-parameter step connecting these practices is less systematically documented for traditional plucked-string instruments, despite prior work on Pipa and Chinese-instrument modeling [7,6,9].

The workflow developed in this paper treats analysis as an initialization and calibration stage rather than as a final resynthesis result. Recordings and partial physical measurements provide constraints for a layered waveguide model in Csound. The goal is not to recover all physical parameters of the instrument from recordings, but to derive physically interpretable control parameters that remain playable. The Csound opcode `wgpluck2` is suitable for this purpose because its controls include pitch, pluck position, damping, and reflection behavior [8,9].

More specifically, this work contributes an analysis-guided workflow for constructing playable Csound physical models from recordings, supplemented by limited physical measurements, demonstrated through rule-based descriptor-to-parameter mapping, a four-layer single-string Tupipa implementation, and diagnostic model evaluation and calibration.

2 Analysis-Guided Model Initialization

Figure 1 summarizes the workflow from controlled recording and measurement to ATS analysis, rule-based descriptor-to-parameter mapping, layered Csound construction, human-in-the-loop calibration, parameter feedback, and final playable-instrument output.

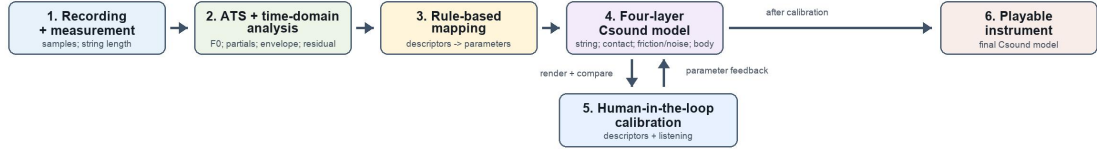


Fig. 1. Analysis-guided workflow for constructing the playable Csound physical model. Calibration feedback is returned to the model parameters before the final instrument is confirmed.

The instrument used in this study is an authentic Liu Yuping Tupipa, with an overall length of 96 cm and a measured effective string length of 65.8 cm. It was selected because Liu is documented as a provincial-level representative inheritor of Tupipa-making craftsmanship in Sichuan’s sixth batch of provincial intangible-cultural-heritage representative inheritors announced in 2016, making the instrument a representative example of locally recognized high-level Tupipa craftsmanship [10]. The recorded techniques followed local practice learned from regional musicians. Controlled recordings consisted of isolated plucks recorded at 44.1 kHz on the same instrument, with microphone placement and plucking position kept consistent across takes. A condenser microphone was placed approximately 15–16 cm horizontally from the plucking point and about 5 cm above the string plane. The right-hand plucking position was approximately 14.5 cm from the bridge, giving a normalized plucking position of $14.5/65.8 = 0.220$. The material included repeated open-string plucks, muted plectrum recordings, and 22 segmented single-string pitch events for each of five conditions: “f” at the regular position, “mf” at the regular position, “p” at the regular position, “mf” near the fingerboard, and “mf” near the nut. These 110 pitch events were recorded as mono 44.1 kHz, 24-bit WAV files.

ATS analysis and time-domain inspection estimated the analyzed note’s fundamental frequency at approximately 369–370 Hz, together with partial structure, attack timing, decay behavior, and residual/noise energy. ATS was selected because its sinusoidal-plus-residual representation preserves harmonic and noise-related temporal behavior in a form compatible with Csound analysis and resynthesis tools [11,12]. In this workflow, ATS therefore supplies harmonic and residual/noise descriptors that are combined with time-domain cues in the subsequent rule-based descriptor-to-parameter mapping.

Table 1. Analysis descriptors and rule-based descriptor-to-parameter mapping.

Analysis descriptor	Target parameter or constraint	Mapping rule
F0	iCps	Set pitch from ATS F0
Amplitude level	kamp	Fit normalized level
String length	pitch plausibility	Check against measured scale
Pluck ratio	iPluckPos	Initialize near 0.220
Partial balance	iPluckPos , damping/filter	Brighter spectra: bridgeward or lower damping
Pick response	kpick	Bounded local sweep of onset brightness
Decay time	krefl	Estimate loss from ATS and envelope
Envelope	gate/release timing	Longer sustain slows release
Attack transient	plectrum-contact envelope	Later peak increases delay
Residual/noise energy	friction/noise band/gain	Set noise-band regions

Table 1 summarizes the practical initialization rules used in this study. Physical measurements provide approximate geometric reference values, while ATS and time-domain descriptors initialize

pitch, level, decay, attack timing, and residual-noise regions. Specifically, `kamp` was fitted to the normalized level, `kpick` was selected by bounded local sweep, and `kref1` was initialized from decay time estimated from ATS partial-amplitude trajectories and the time-domain envelope. These rules define initial values and plausible ranges; final parameter choices are refined later during model evaluation and calibration.

3 Layered Csound Instrument

The initialized parameters are then used to construct a four-layer Csound instrument (Fig. 2). The proposed single-string model is organized into core string-vibration, plectrum-contact, friction/noise, and body-response layers. This layered architecture follows the analysis stage: ATS and time-domain inspection identify pitched harmonic sustain, short plectrum-contact transients, residual friction/noise energy, and body-response spectral shaping as separate but interacting synthesis layers.

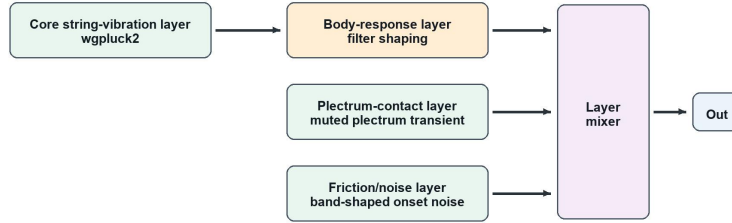


Fig. 2. Four-layer Csound instrument architecture. The body-response layer filters the core string-vibration signal, while the plectrum-contact and friction/noise layers contribute short onset components before final mixing.

The core string-vibration layer produces the pitched string response with `wgpluck2`. Its `iCps` value is initialized from ATS `F0`, `iPluckPos` from the measured 14.5/65.8 ratio, and `kamp`, `kpick`, and `kref1` from the level, onset, and decay rules summarized in Table 1. After calibration, the final setting used `iCps`=369.0 Hz, `kamp`=0.30, `iPluckPos`=0.220, `kpick`=0.400, and `kref1`=0.624. The confirmed implementation used core gain 0.29, body-response gain 0.78, plectrum-contact gain 0.005, friction/noise gain 0.11, core high-pass filtering at 105 Hz, and core low-pass filtering at 7100 Hz. At performance time, an optional score `p4` value in Hz replaces the 369-Hz default. The selected frequency is passed to `wgpluck2`, the explicit partial oscillators, the pitch-dependent body bands, and the tied resonance, so the same instrument can be rendered at the measured single-string pitch targets.

The plectrum-contact layer represents the short onset color associated with plectrum-string contact. Muted-string recordings provide a reference for its timing and spectral color, and the layer is implemented as a controllable onset component. The four custom opcodes used in the implementation are study-specific UDOs (user-defined opcodes), not external plug-ins: `TupipaBodyWg2` applies the body and harmonic filters, `PlectrumContactWg2` shapes the muted-plectrum transient, `FrictionNoiseWg2` creates the band-limited friction component, and `LowBodyTransientWg2` provides an optional low-frequency body-transient path. Their complete definitions, comments, input signatures, and render settings are included in the complete commented Csound source supplied by the authors. The core signal is explicitly high-pass filtered at 105 Hz and low-pass filtered at 7100 Hz in that source file.

The friction/noise layer represents short, band-limited noise associated with plectrum friction. It uses band-shaped noise controlled by the onset envelope, with initial frequency regions set from residual/noise-energy inspection and muted plectrum recordings.

The body-response layer approximates the instrument-body coloration as broad spectral shaping. Its filter settings are calibrated to reduce excessive high-frequency energy and match the target spectral balance.

For reproducibility, the complete commented Csound source and the listening report are available at the permanent supplementary-materials URL provided by the authors: https://drive.google.com/file/d/12_A5xcZU9KIci6Eacglz0yhm1Zbg3Mdf/view?usp=sharing.

4 Model Evaluation and Calibration

Evaluation consisted of three stages. First, the ATS reconstruction and the playable physical model were compared against the target recording. Second, acoustic descriptors were used diagnostically to assess attack behavior and spectral similarity. Third, selected controls such as `kpick`, `krefl`, contact gain, and friction/noise gain were swept within plausible ranges before final refinement. The ATS reconstruction served as an acoustic reference because it approximates measured spectral detail without exposing physically interpretable control parameters. Since the goal is parameter initialization rather than codec evaluation, four descriptors were used for diagnostic comparison: onset envelope for attack timing and shape, spectral centroid for brightness, log-spectral distance for full-band spectral deviation, and log-mel distance for perceptually compressed spectral shape. Descriptor values were computed after onset alignment and amplitude normalization. Full-window descriptors are used for relative comparison of global spectral shape. Because the plectrum-contact and friction/noise layers are short onset components, their effects are concentrated in the attack stage and are therefore inspected in the 0–80 ms attack window.

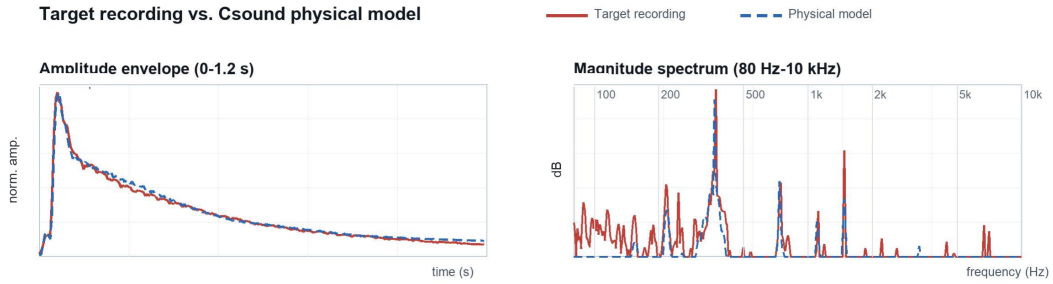


Fig. 3. Diagnostic comparison between the target recording and the calibrated physical model. The envelope and spectrum plots were used for iterative parameter refinement rather than objective evaluation.

Using the same alignment and amplitude normalization, the complete model yielded a log-spectral distance of 23.97 dB over the full 1.2-s note and 31.31 dB in the 0–80-ms attack window. The two values describe different aspects of the comparison: overall spectral similarity and attack-transient similarity, respectively.

To address pitch-range coverage, we analyzed all 110 segmented single-string events described above. Across the five conditions, the detected F0 estimates span 247.8–918.8 Hz. The set contains 28 distinct measured frequency targets after retaining small condition-dependent tuning differences. We rendered the complete model at each target through the optional `p4` input. All 28 renders completed with zero Csound performance errors and no clipping; the largest peak across the rendered set was -4.85 dBFS and the mean DC offset was approximately -1.5×10^{-5} . Spectral inspection found the requested fundamental or its harmonic series in every render. The test therefore confirms pitch control across the recorded single-string range at the calibrated reference timbre, while not claiming an independently retuned spectral envelope at every pitch.

The balanced control subset also retained five repeated plucks in each of “f”, “mf”, and “p”, with median peak amplitudes of 0.657, 0.361, and 0.057 and median 20-dB decay times of 0.375, 0.450, and 0.422 s, respectively. A second controlled set at 3.0, 14.5, and 24.5 cm from the bridge (normalized positions 0.046, 0.220, and 0.372) yielded median attack spectral centroids of 771, 455, and 406 Hz. Together, the pitch, intensity, and plucking-position tests provide compact evidence for a playable single-string model.

Calibration used the mapping rules in Table 1 to define plausible parameter ranges. Within those ranges, bounded local parameter adjustment was carried out around selected controls, including

`kpick`, `krefl`, contact gain, and friction/noise gain. Acoustic-descriptor differences relative to the target recording were used to reject implausible settings, after which final parameter refinement was guided by an informal listening group consisting of the author, two conservatory professors specializing in Chinese traditional music and sound design, and experienced local Pipa performers and cultural practitioners. The group compared the target, complete model, and isolated layer renders and discussed attack articulation, brightness, body coloration, friction, and sustain. The retained record is qualitative: the group preferred a moderate rather than sharply bright attack, a distinct but subordinate plectrum transient, restrained friction noise, and a longer body-supported decay. Individual ratings, session date, and playback hardware were not retained, so no inter-rater agreement or statistical listening result is claimed. The panel therefore served as a calibration check, not as a standalone perceptual experiment. This process was not an automatic global optimizer; it uses acoustic descriptors to guide calibration while preserving human-in-the-loop judgment.

5 Discussion and Conclusion

The case study shows that acoustic descriptors narrow plausible parameter ranges while listening remains necessary for final calibration. The recorded single-string material covers a detected F0 span of 247.8–918.8 Hz, and the 28-target render test confirms that the Csound instrument can produce every measured pitch in that recorded span without render errors or clipping. The optional performance-time pitch input makes this a playable single-string model with complete recorded-range pitch coverage, rather than a model restricted to the 369-Hz reference note. The intensity and plucking-position tests further show that the model is grounded in control-relevant performance variation, although its timbral calibration remains anchored to the reference condition. The next development step is to model the unison double course and its coupled string vibration, followed by other performance techniques such as articulated damping, slides, bends, and tremolo.

References

1. He, Z.: The Inquiry of Formation and Development and Innovation about Sichuan Jiuzhaigou County's Nanping Tune (in Chinese). Master's thesis, Xiamen University (2017)
2. CPPCC Jiuzhaigou County Committee (ed.): Jiuzhaigou County Folk Song Culture. Jiuzhaigou County Historical Materials, vol. 5 (in Chinese). CPPCC Jiuzhaigou County Committee, Jiuzhaigou (2004)
3. Serra, X., Smith, J.O.: Spectral modeling synthesis: A sound analysis/synthesis system based on a deterministic plus stochastic decomposition. *Computer Music Journal* 14(4), 12–24 (1990)
4. Pampin, J.: ATS: A system for sound analysis, transformation, and synthesis based on a sinusoidal plus critical-band noise model (2004)
5. Jaffe, D.A., Smith, J.O.: Extensions of the Karplus-Strong plucked-string algorithm. *Computer Music Journal* 7(2), 56–69 (1983)
6. Smith, J.O.: Acoustic modeling using digital waveguides. In: Roads, C., Pope, S.T., Piccialli, A., De Poli, G. (eds.) *Musical Signal Processing*, pp. 221–263. Swets and Zeitlinger, Amsterdam (1997)
7. Chen, Y.-H., Huang, C.-F.: Sound synthesis of the Pipa based on computed timbre analysis and physical modeling. *IEEE Journal of Selected Topics in Signal Processing* 5(6), 1170–1179 (2011)
8. Csound: `wgpluck2`. Csound Manual, <https://csound.com/manual/opcodes/wgpluck2/>
9. Boulanger, R. (ed.): *The Csound Book: Perspectives in Software Synthesis, Sound Design, Signal Processing, and Programming*. MIT Press, Cambridge (2000)
10. Sichuan Intangible Cultural Heritage Protection Center: Official representative-inheritor information interface (in Chinese), <https://ichsc.tsichuan.com/#/inheritor>, accessed 14 September 2026
11. Csound: `ATSadd`. Csound Manual, <https://csound.com/docs/manual/ATSadd.html>
12. Csound: `ATSinnoi`. Csound Manual, <https://csound.com/docs/manual/ATSinnoi.html>